

SECOND PUBLIC EXAMINATION

Honour School of Philosophy, Politics and Economics

Honour School of Economics and Management

Honour School of History and Economics

Honour School of Engineering, Economics and Management (Part II)

Honour School of Materials, Economics and Management (Part I)

MICROECONOMICS

TRINITY TERM 2015

Friday 29th May 2015, 14:30 - 17:30

Please start the answer to each question on a separate page.

There are 11 questions in this paper.

*Answer **three** short questions from Part A and **two** long questions from Part B.*

Part A attracts 1/3 of the marks in total; each question in part B attracts 1/3 of the marks.

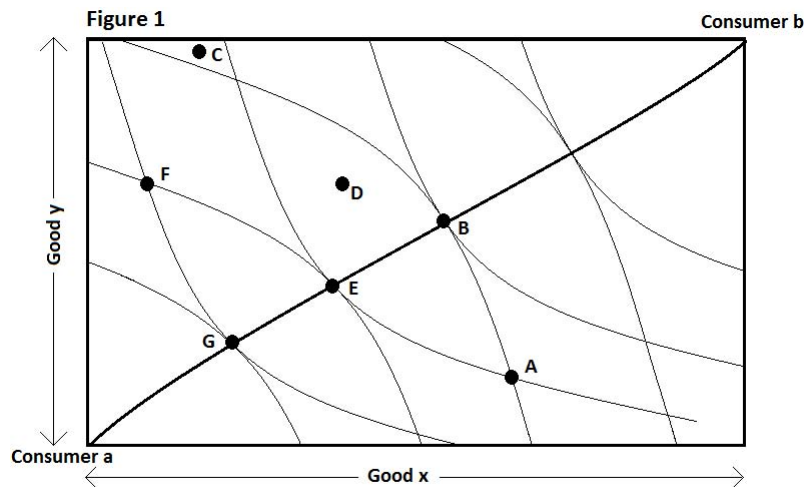
For multipart questions, the weight assigned to each part is indicated in square brackets.

Candidates may use their own calculators.

Do not turn over until told that you may do so.

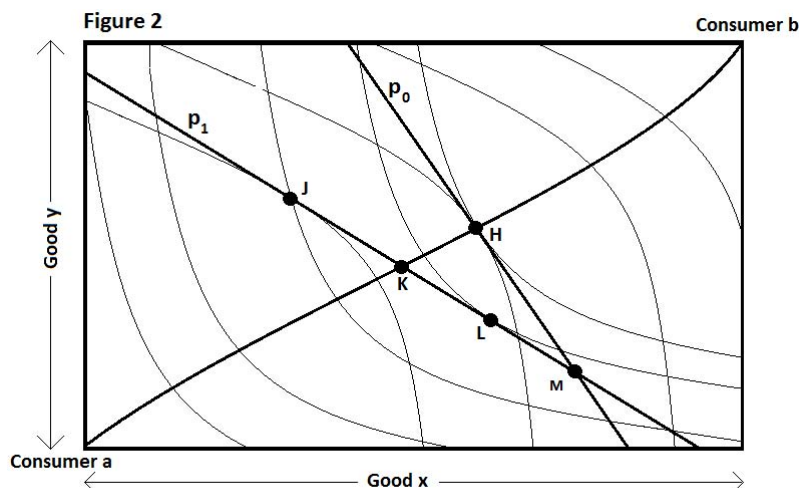
Part A

1. (a) Explain the Pareto Criterion for choosing between social outcomes. What are its advantages and disadvantages? [15%]



The Edgeworth Box in Figure 1 represents all possible allocations between consumers a and b of two goods x and y in an exchange economy. The consumers have convex preferences. Indifference curves and the contract curve are shown.

- (b) Which of the labelled points in Figure 1 are Pareto efficient? Which are Pareto superior to point A? Which are Pareto inferior to A? [20%]
- (c) “Point G is Pareto efficient and Point A is not, but G is not Pareto superior to A.” Is this statement true? Explain your answer. Suggest a criterion that a social planner could use to choose between A and G. [20%]



In Figure 2, M is the initial endowment. Let the price of good y be 1. The straight lines correspond to two possible prices p_0 and p_1 for good x .

- (d) Explain carefully why p_0 represents a competitive equilibrium price ratio, and p_1 does not. Use the terms *excess demand* and *excess supply* in your answer. [25%]
- (e) Suppose that the social planner prefers point K to point H, and is able to redistribute good x but not good y . Explain what might be done to achieve allocation K. Would the equilibrium price ratio be greater or less than p_1 ? [20%]
2. Suppose that two firms produce horizontally differentiated products. The firms compete for a single period and choose prices simultaneously. Each firm has the same constant marginal cost of production $c \geq 0$, and fixed costs are zero. The linear demand functions are given by:

$$\begin{aligned} q_1(p_1, p_2) &= 100 - \alpha p_1 + \beta p_2 \\ \text{and } q_2(p_1, p_2) &= 100 - \alpha p_2 + \beta p_1. \end{aligned}$$

Assume that $\alpha > \beta > 0$ and that $100 > \alpha c$.

- (a) Explain the difference between horizontally differentiated and vertically differentiated products, giving examples of each. Explain why the demand functions above represent horizontally differentiated products. [25%]
- (b) Show that the symmetric pure-strategy Bertrand-Nash equilibrium prices are:

$$p_i^* = p_j^* = \frac{100 + \alpha c}{2\alpha - \beta}$$

Find the corresponding quantities, and check that the prices and quantities are positive at the equilibrium. [25%]

- (c) Show how the equilibrium prices change in the parameters α, β and c . Provide a brief economic explanation of these comparative statics results. [25%]
- (d) Suppose now that the two firms merge. The new firm acts as a monopolist, choosing p_1 and p_2 . Find the profit-maximizing price assuming that the monopolist sets the same price for both products, that is $p_1 = p_2$. Is this price higher or lower than the equilibrium price from part (b)? [25%]

3. An agent can exert two effort levels, $e = 1$ or 0 . When $e = 1$ a production error occurs with probability 0.25; when $e = 0$ the probability of a production error is 0.75. His utility is:

$$u(w, e) = 4 - \frac{5}{w} - 2e,$$

where w is the wage received. Production errors are observable and so can be introduced into the agent's contract, but effort cannot. The output produced is worth 12 if there is no error and 0 otherwise. The principal is risk neutral. Assume the agent's reservation utility is $\underline{u} = 0$.

- (a) Suppose the principal wishes the agent to undertake effort $e = 0$. What is the participation constraint in this case? What can you say about the incentive constraint? Find the wage schedule she will offer the agent, conditional on the observed error. What is the principal's expected profit? [35%]
 - (b) Suppose the principal wishes the agent to undertake effort $e = 1$. Write down the participation and incentive constraints. Find the wage schedule she will offer the agent, conditional on the observed error. What is the principal's expected profit in this case? [45%]
 - (c) Find the agency cost involved in implementing effort $e = 1$. [20%]
4. Arthur is a risk-averse farmer whose income will be y if the weather is fine during harvest, and $y - L$ if it rains. The probability of rain is 0.5.
- (a) Draw a carefully-labelled 2-state diagram to illustrate: his endowment point; his expected income; his certainty-equivalent income. [25%]
 - (b) Use the diagram to illustrate and explain (i) how he would benefit from buying insurance in a competitive insurance market; (ii) the maximum expected profit that his risk-neutral neighbour Bertha could make by offering him full insurance. [25%]
 - (c) Suppose instead that Arthur was risk-seeking. Describe, and illustrate on a 2-state diagram, an agreement with Bertha that would make Arthur better off, and that Bertha would be just willing to accept. [20%]

Now suppose Arthur and Bertha are both risk-averse farmers with utility of income, y , given by $u(y) = \ln y$. On each farm, income will be 8 if it is fine, and 2 if it rains.

As before, the probability of rain on each farm is 0.5. Furthermore, with probability $p \leq 0.5$, both will have fine weather, with probability p it will rain on both farms, and otherwise just one of them will have rain.

- (d) Find the increase in utility that each of them could achieve by pooling their incomes, as a function of p . How do the benefits of pooling change as p increases? Explain intuitively why this happens. [30%]

5. (a) What is the *tragedy of the commons*? Use the concepts of rivalry and excludability in your answer. [20%]

A particular road is used by N drivers each day. Each driver obtains a benefit v from the journey, and incurs a cost $c(N)$ that increases with the number of drivers on the road, because of congestion. The cost incurred by each driver is given by:

$$c(N) = a + bN$$

- (b) Show that the equilibrium number of drivers using the road will be:

$$N_e = \frac{v - a}{b} \quad [10\%]$$

- (c) Write down an expression for the social surplus, W (total benefits minus costs), provided by the road when it is used by N drivers. Find the number of drivers, N^* , at which the social surplus is maximised. What is the social surplus when $N = N_e$? Comment on your answers. [25%]
- (d) Suppose the government charges a toll, t , (a congestion charge) for each journey. Find the optimum toll. Interpret your answer in terms of the externality that each driver imposes on others. [25%]

By spending an amount X on road improvements, the government reduces the cost per trip to:

$$c(N, X) = a + bN - gX$$

Assume there is no congestion charge.

- (e) Write down the new expression for the social surplus, W (including the cost, X , incurred by the government), when there are N drivers. Find the new equilibrium traffic flow, N_e , and the equilibrium social surplus. Comment on the social value of road improvements. [20%]

Part B

6. “When markets are competitive, buyers and sellers don’t need to know anything about each other. As long as they know the market prices, the equilibrium will be efficient.”

“When there is asymmetric information between buyers and sellers, market failure is the likely result.”

Discuss these two claims. How can they be reconciled?

7. Explain the Samuelson condition for the efficient provision of a public good. Evaluate the following alternative proposals for funding the BBC:
- (a) A broadcast levy, to be paid by all households, regardless of whether they watch television.
 - (b) Pay-per-view, where consumers pay for each programme watched.
 - (c) Subscriptions, where a subscription provides access to a particular channel.
 - (d) A licence fee, as at present, paid by all households where television is watched.

Are there any economic reasons to think that the alternative proposals might result in different types of TV programmes?

8. When a competition authority considers the possible effects of a merger, how can it assess: (i) any increase in market power that might arise from the merger; and (ii) any cost efficiencies that might arise from the merger? Should a competition authority ever permit a merger if it increases market power?
9. Using the concept of subgame perfection, explain how firms might be able to collude in a market. How would you expect the number of firms in the market to affect their ability to collude successfully?
10. Pat argues, “Sure, I don’t like risk. But if I always choose the option with the higher expected value, over the long run the wins and losses will more or less cancel out, except that I will end up getting a higher payoff on average. So it is irrational to be averse to risk.” Evaluate Pat’s argument.

11. An economy ('Home') has a fixed endowment of labour, L , which it can employ in two sectors with outputs x and y . The sectors have production functions:

$$x = L_x^{1/2}, \quad y = L_y^{1/2}$$

where L_x and L_y are the labour inputs in the two sectors, and $L_x + L_y = L$.

- (a) Derive, and sketch, the production possibility frontier of the economy. What is the marginal rate of transformation between x and y ? [15%]

The economy contains a single consumer with utility function $U(x, y) = x^\alpha y^{1-\alpha}$, where $0 < \alpha < 1/2$. The consumer receives the total income of the economy, in the form of wages and profits.

- (b) What is the marginal rate of substitution between x and y ? [5%]
 (c) Let the price of good y be 1. Find the competitive equilibrium price, p , of good x , and the equilibrium quantities of the two goods. Explain how and why the quantities produced depend on the parameter α . Illustrate the equilibrium on your diagram. [30%]

Another economy ('Foreign') has the same technology, and but it has labour endowment L^* , and its consumer has a (different) utility function $U(x, y) = x^{1-\alpha} y^\alpha$. The two countries trade freely.

Suppose that $L = L^*$.

- (d) Show that the world price ratio, p , must be 1, explaining your reasoning carefully. [15%]
 (e) What is the pattern of trade between the countries? [10%]
 (f) Are there gains from trade? Explain and illustrate your reasoning carefully. (A formal proof is not required.) [10%]
 (g) What happens to the world price ratio, p , if L^* increases? What happens to real income in the Home country? [15%]